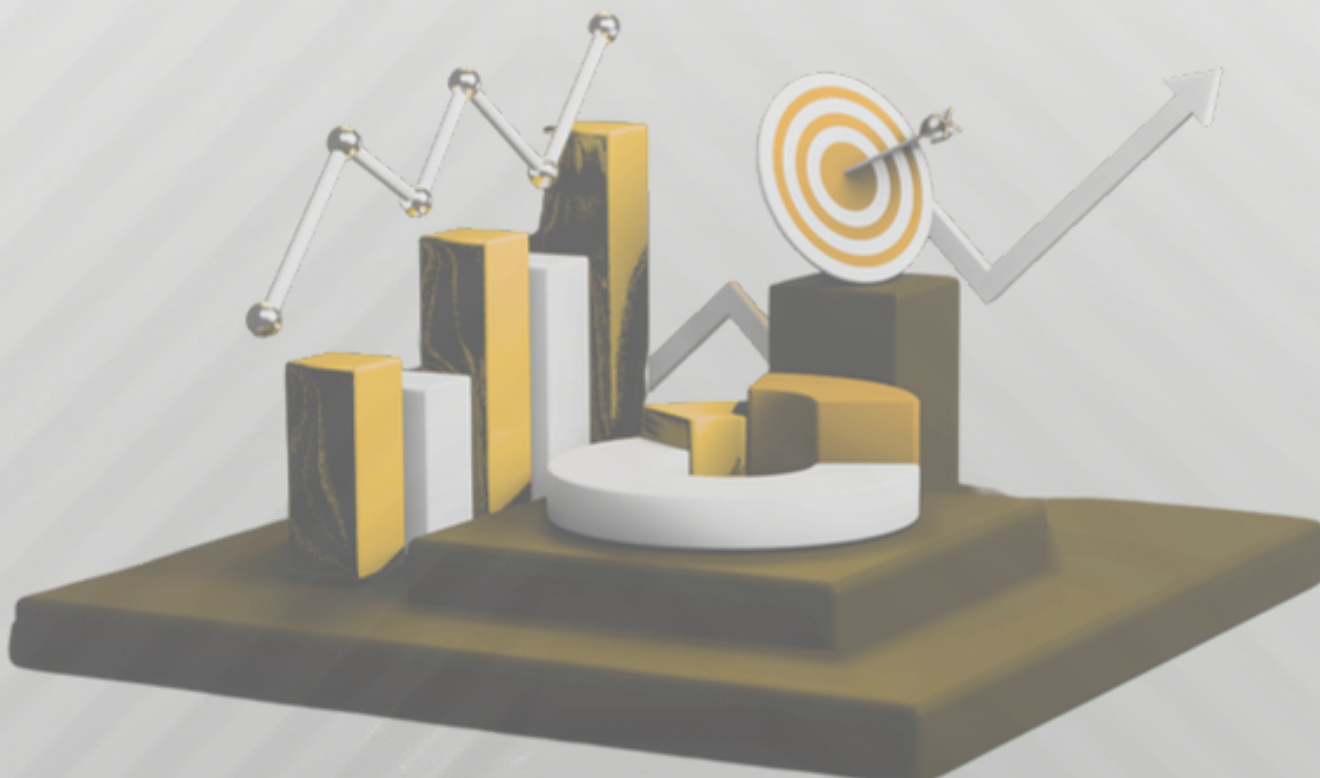




Visualizing Data with



pythonTM



Python Libraries for Data Visualization

Matplotlib: The foundational plotting library, used for basic charts.

Seaborn: Built on top of Matplotlib, ideal for statistical plots.

Plotly: Used for interactive and web-based visualizations.

< matplotlib >

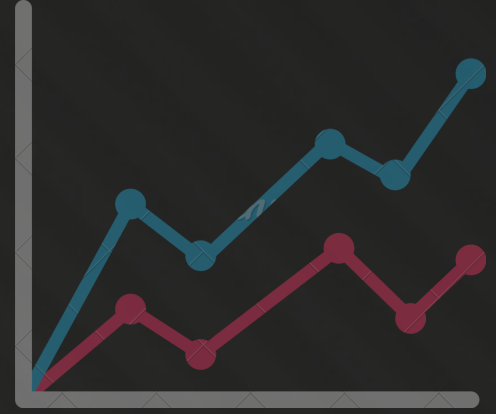


seaborn



plotly

Line Plots



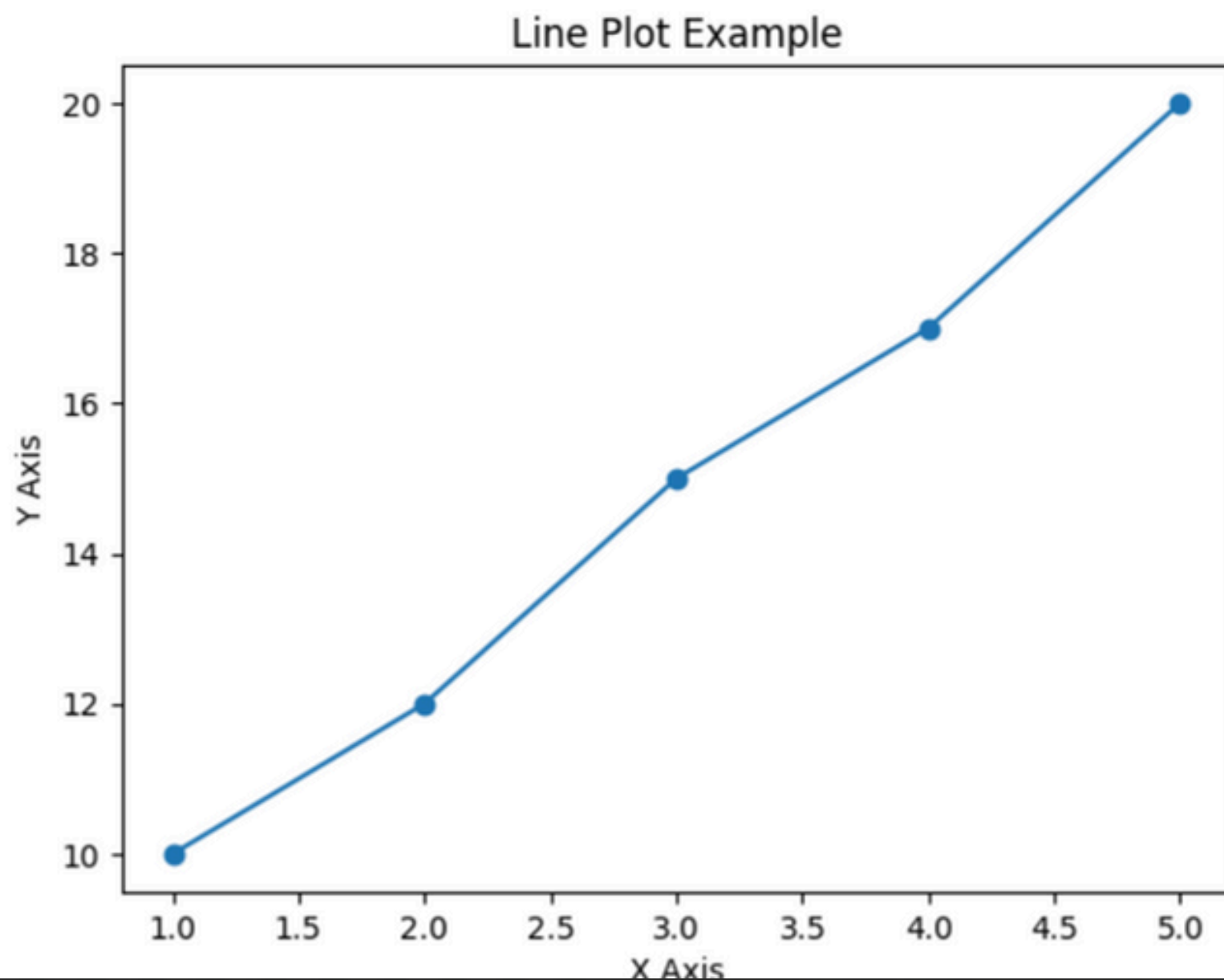
Use Case: Showing trends over time.

Library: Matplotlib

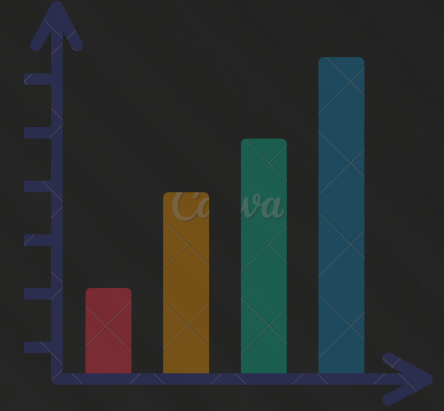
```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
x = [1, 2, 3, 4, 5] # X-axis data
y = [10, 12, 15, 17, 20] # Y-axis data

# Create Line Plot
plt.plot(x, y, marker='o') # Plot x and y with circle markers on each point
plt.title('Line Plot Example') # Add a title to the plot
plt.xlabel('X Axis') # Label for the X-axis
plt.ylabel('Y Axis') # Label for the Y-axis
plt.show() # Display the plot
```



Bar Plots



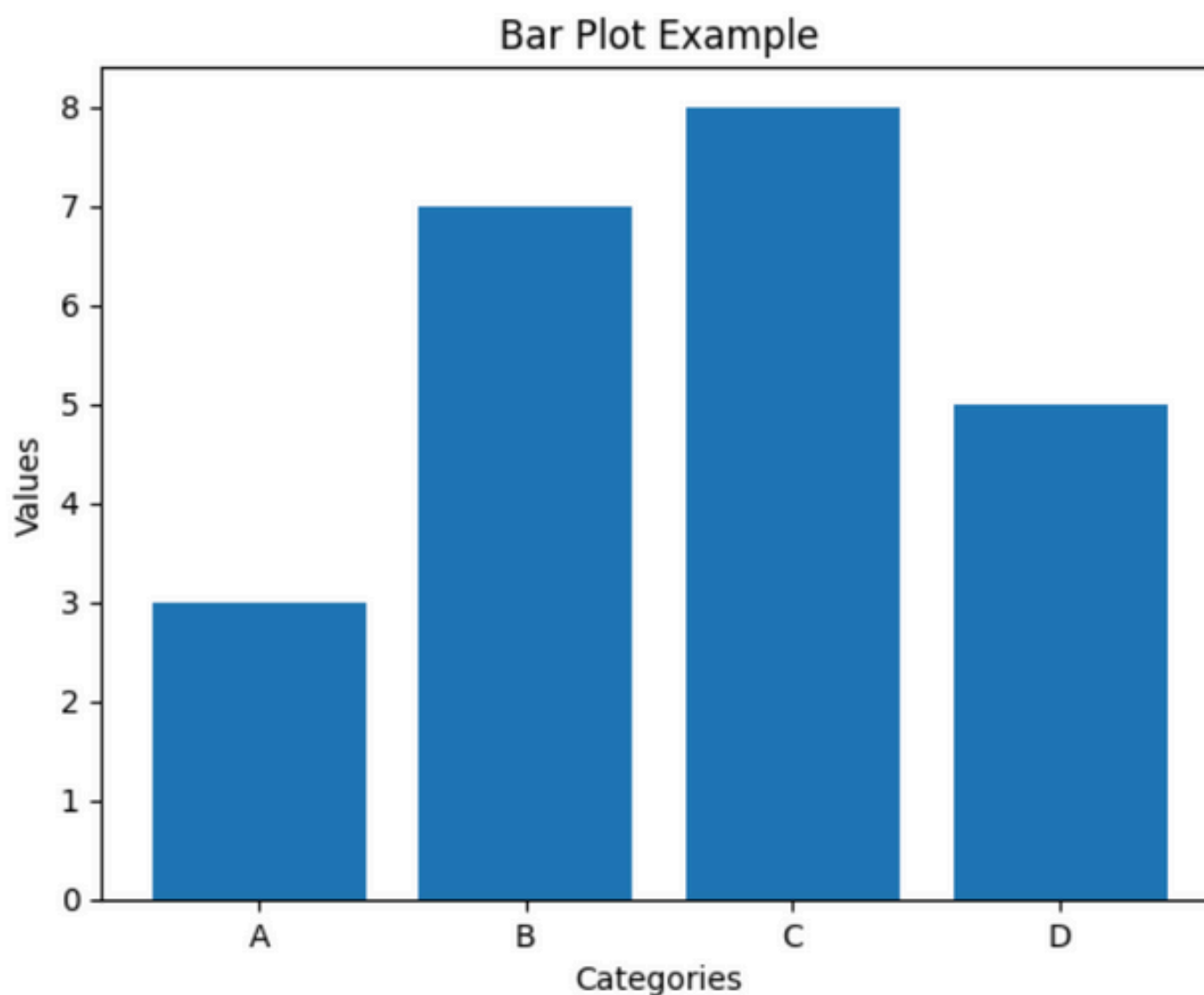
Use Case: Comparing different categories or discrete items.

Library: Matplotlib

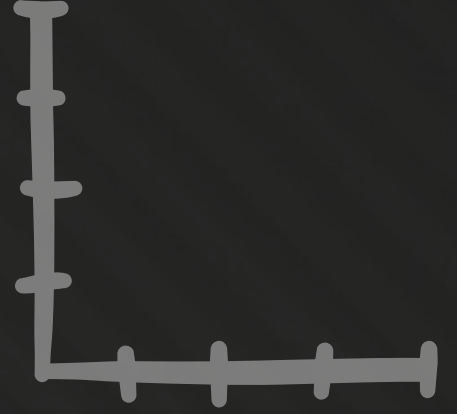
```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
categories = ['A', 'B', 'C', 'D'] # Define categories
values = [3, 7, 8, 5] # Corresponding values for each category

# Create Bar Plot
plt.bar(categories, values) # Create a bar plot using categories and values
plt.title('Bar Plot Example') # Add a title to the plot
plt.xlabel('Categories') # Label for the X-axis
plt.ylabel('Values') # Label for the Y-axis
plt.show() # Display the plot
```



Scatter Plots



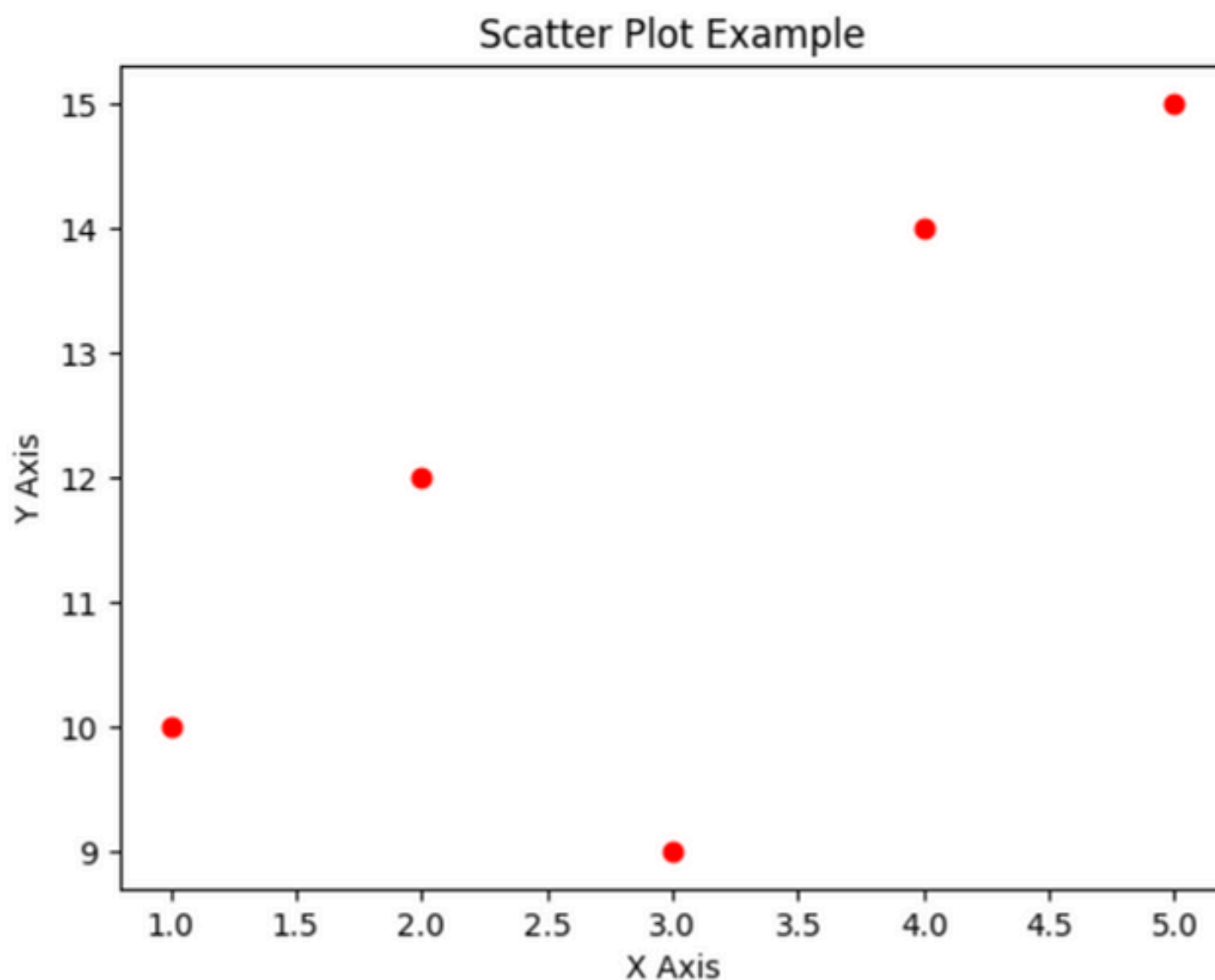
Use Case: Showing relationships between two variables.

Library: Matplotlib

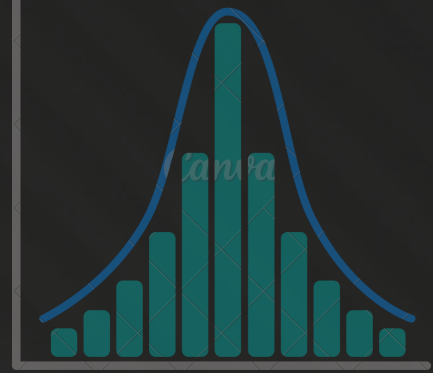
```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
x = [1, 2, 3, 4, 5] # X-axis data
y = [10, 12, 9, 14, 15] # Y-axis data

# Create Scatter Plot
plt.scatter(x, y, color='red') # Create a scatter plot with red dots
plt.title('Scatter Plot Example') # Add a title to the plot
plt.xlabel('X Axis') # Label for the X-axis
plt.ylabel('Y Axis') # Label for the Y-axis
plt.show() # Display the plot
```



Histograms



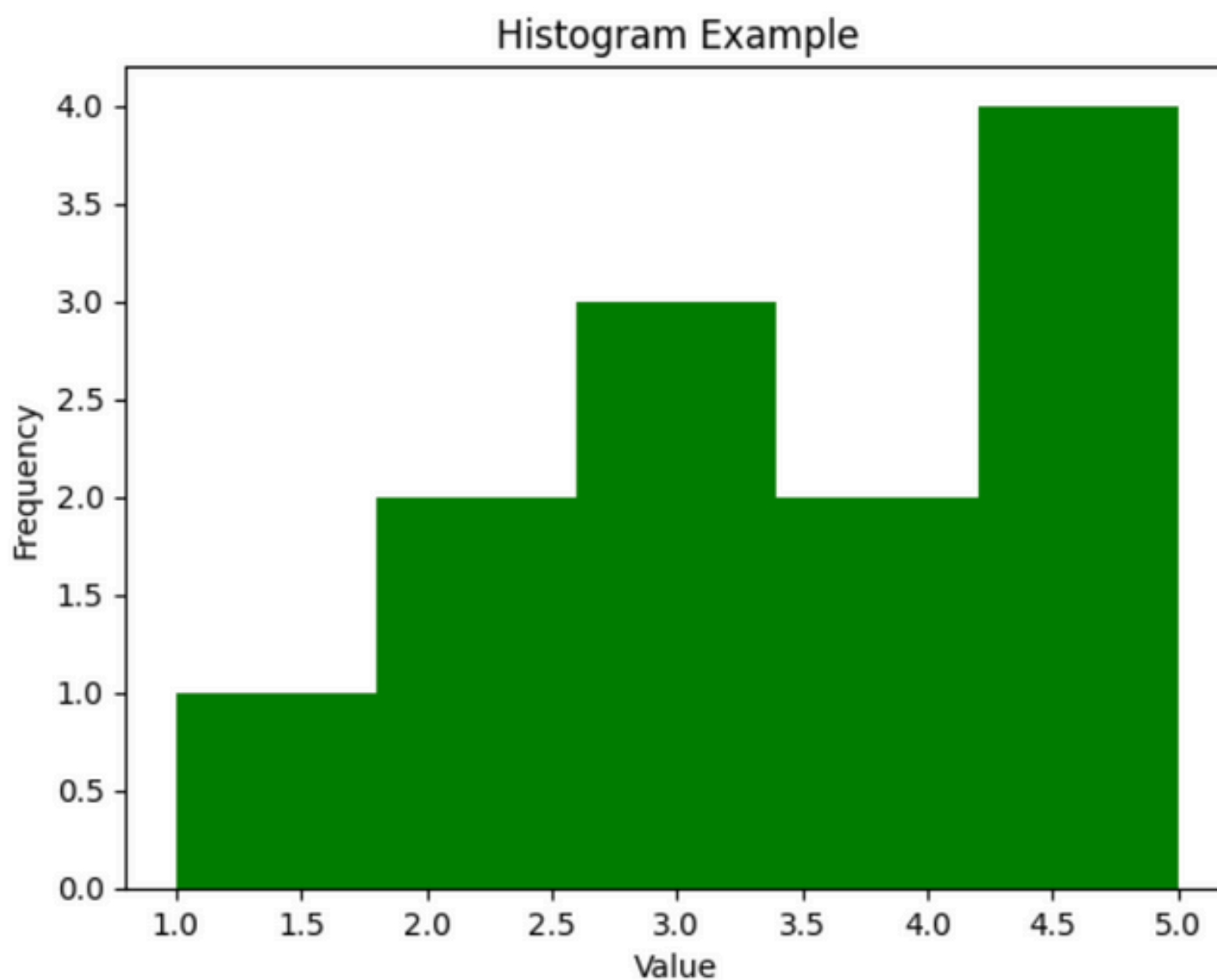
Use Case: Showing the distribution of data.

Library: Matplotlib

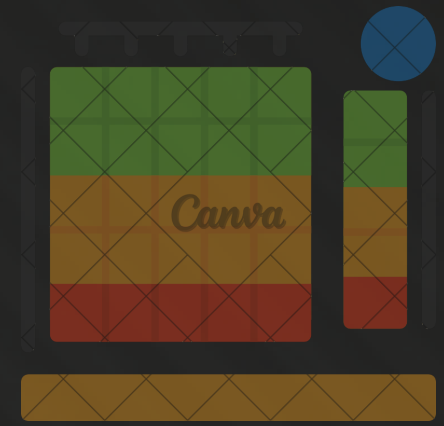
```
import matplotlib.pyplot as plt # Import the Matplotlib library

# Data
data = [1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 5] # Sample data to plot

# Create Histogram
plt.hist(data, bins=5, color='green') # Create a histogram with 5 bins and green bars
plt.title('Histogram Example') # Add a title to the plot
plt.xlabel('Value') # Label for the X-axis
plt.ylabel('Frequency') # Label for the Y-axis (frequency of each bin)
plt.show() # Display the plot
```



Heatmaps



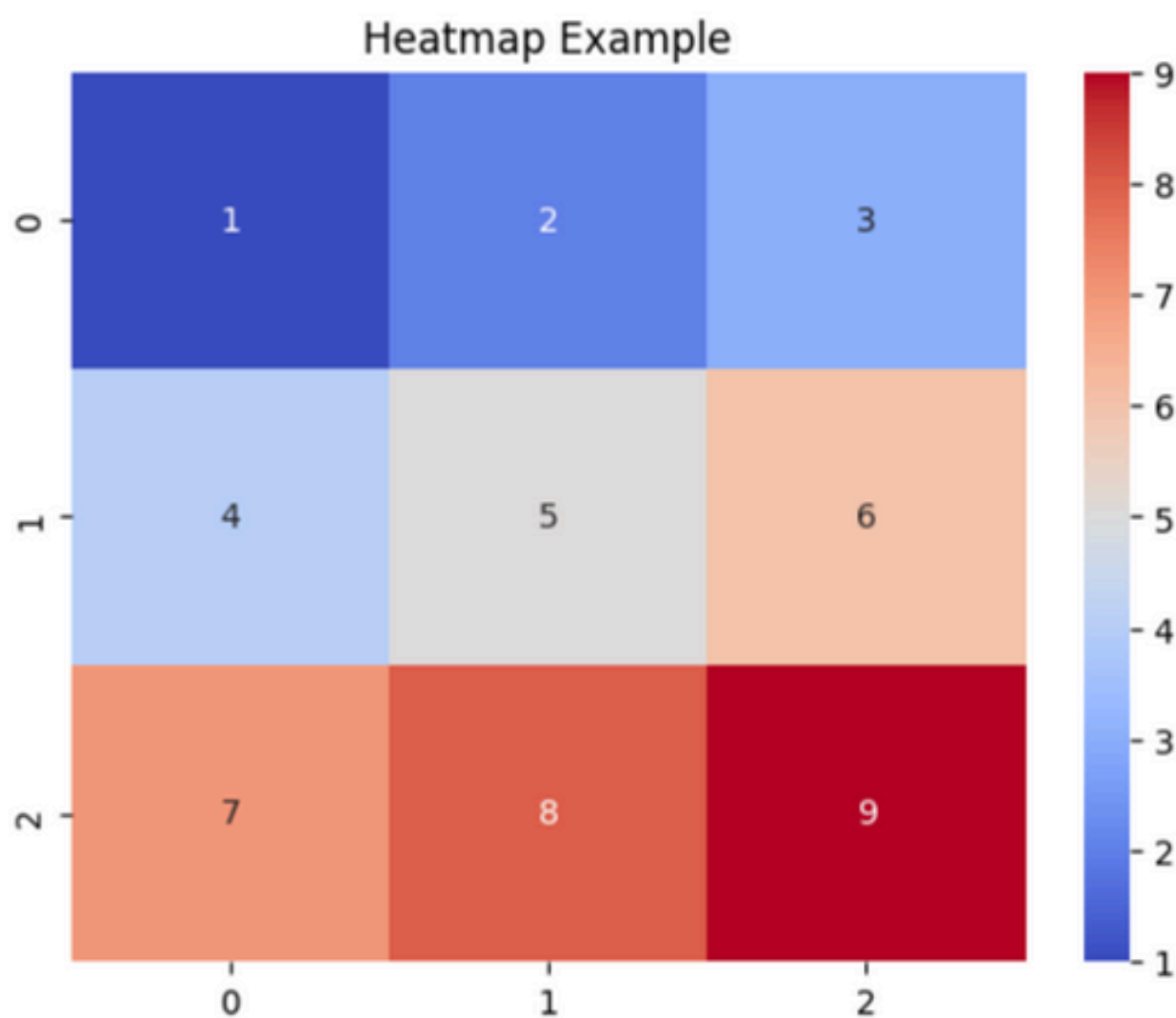
Use Case: Visualizing data in matrix form (correlation, confusion matrix).

Library: Seaborn

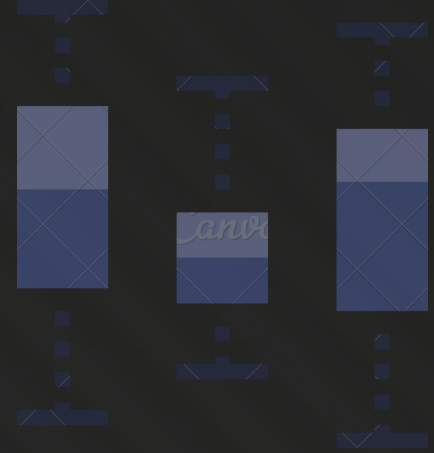
```
import seaborn as sns # Import the Seaborn library
import matplotlib.pyplot as plt # Import Matplotlib for additional control

# Data
data = [[1, 2, 3], [4, 5, 6], [7, 8, 9]] # Sample 2D array to visualize as a heatmap

# Create Heatmap
sns.heatmap(data, annot=True, cmap='coolwarm') # Create heatmap with annotations and color map
plt.title('Heatmap Example') # Add a title to the heatmap
plt.show() # Display the heatmap
```



Box Plots



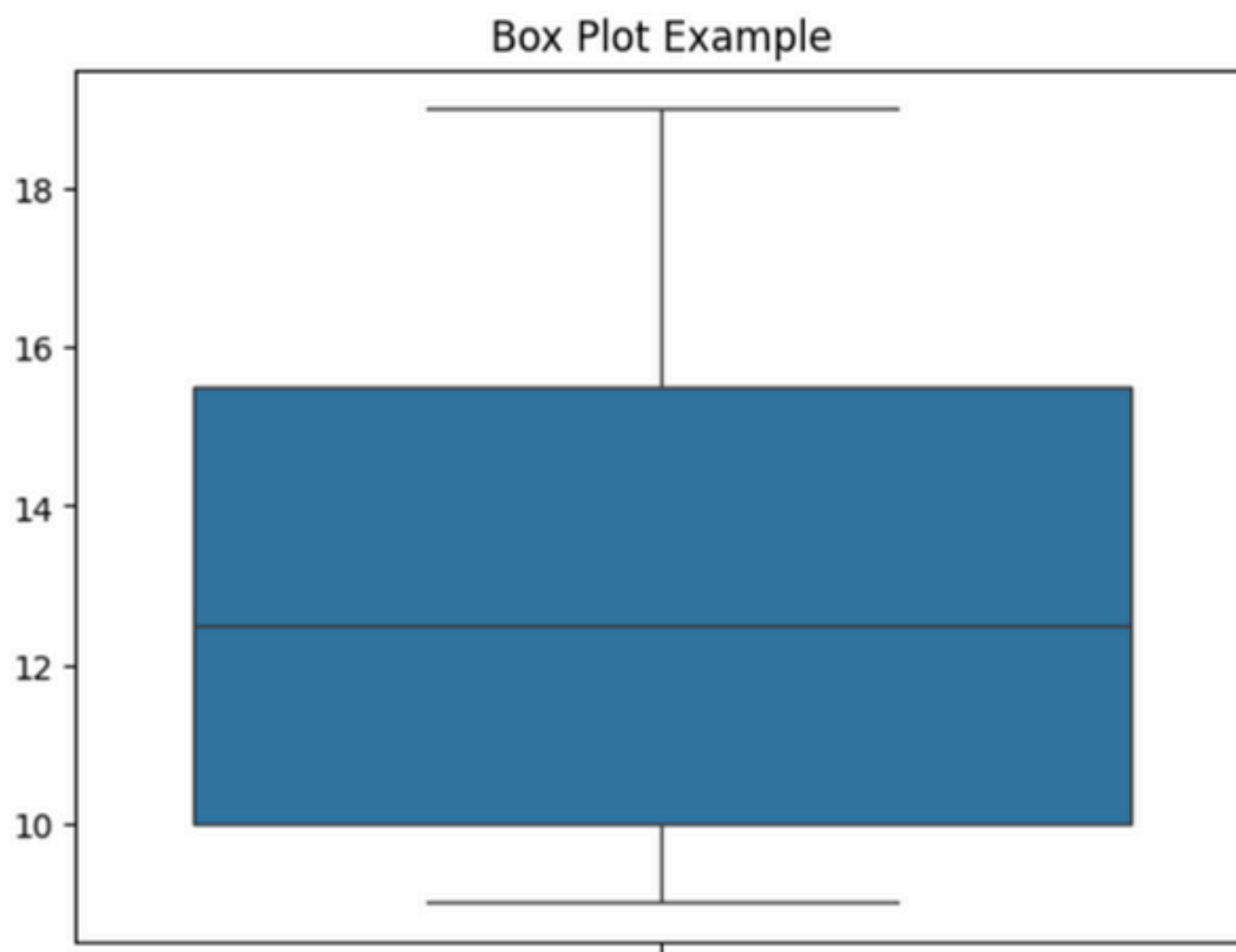
Use Case: Summarizing the distribution of a dataset, showing quartiles and outliers.

Library: Seaborn

```
import seaborn as sns # Import the Seaborn library
import matplotlib.pyplot as plt # Import Matplotlib for additional control

# Data
data = [10, 12, 15, 13, 17, 19, 10, 9] # Sample data for the box plot

# Create Box Plot
sns.boxplot(data=data) # Create a box plot to show distribution, quartiles, and outliers
plt.title('Box Plot Example') # Add a title to the plot
plt.show() # Display the plot
```



Interactive Plots



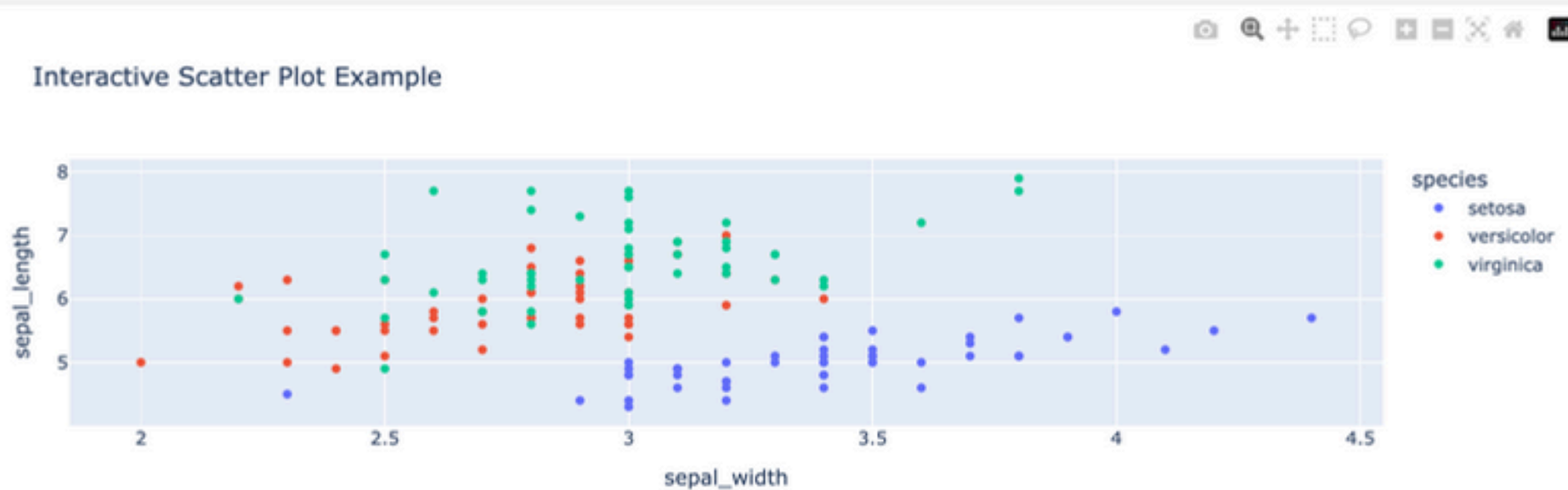
Use Case: Interactive plots for dashboards, complex analysis.

Library: Plotly

```
import plotly.express as px # Import Plotly Express for simpler interactive plotting

# Sample Data
df = px.data.iris() # Load built-in Iris dataset

# Create Interactive Scatter Plot
fig = px.scatter(df, x='sepal_width', y='sepal_length', color='species', # Create an interactive scatter plot
                title="Interactive Scatter Plot Example") # Add a title to the plot
fig.show() # Display the interactive plot
```



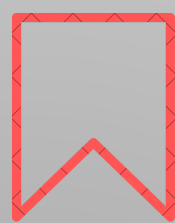
Conclusion

Recap: Each visualization serves a specific purpose; choose the one that best fits your data.

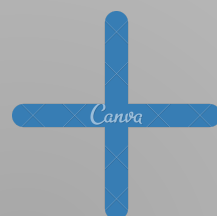
Best Practices:

- Always include labels and titles.
- Use color effectively.
- Consider interactive plots for reports or dashboards.
- Use combination charts when necessary (e.g., bar and line together).

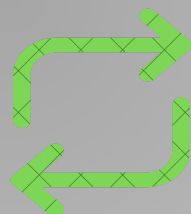
FOUND THIS USEFUL?



SAVE IT!



FOLLOW ME



REPOST IT

